WHITE PAPER · HWACHEON MEGA 95 × AEROSPACE

Hwacheon Mega 95 for Aerospace

Long-bed CNC lathe for extended shafts, rollers, rods, and structural round parts. This paper covers what the machine does, the aerospace parts we produce with it, the alloys we run, the tolerances and finishes aerospace buyers expect, and the documentation package.

16" OD × 120" L max turning
ENVELOPE

±0.0005" routine on critical features
TOLERANCE

1994
SINCE

Same-week / rig-down direct
RESPONSE



Rattle & Hum · 16" OD × 120" L max turning · long-bed heavy-duty CNC lathe

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Executive summary

Machine: Hwacheon Mega 95 (shop nickname: *Rattle & Hum*). long-bed heavy-duty CNC lathe at B&R Productions, running from our New Waverly, Texas shop.

Application: Long-bed CNC lathe for extended shafts, rollers, rods, and structural round parts for Aerospace customers — Aerospace primes, tier-1 and tier-2 suppliers, MRO shops, propulsion component manufacturers, structural assembly integrators, actuator OEMs.

Envelope: 16" OD × 120" L max turning. **Tolerance:** ±0.0005" routine on critical features. **Traceability:** Material by heat and lot on every job.

The machine, in context

Ten feet of turning length is a specialty capability. Long drive shafts, extended mandrels, rig-floor rollers, oil-and-gas structural rods — the parts that don't fit on a 4- or 6-foot lathe. Rattle & Hum lets us keep this class of work in-house instead of outsourcing the finish turning after roughing on a larger machine somewhere else.

Why Rattle & Hum for Aerospace work

Long aerospace round parts — shafting, actuator rods, long structural rounds — up to 120 inches in a single piece. The value is the same as it is everywhere long parts matter: one piece means no joint, no coupling, and no tolerance stack between sections.

What this looks like on a real part

Straightness governs at this length, and on an aerospace part it is usually an inspected characteristic rather than an assumed one. The part deflects under its own weight, deflects again under tool pressure, and grows measurably as roughing heat goes in. Roughing and finishing get treated as separate thermal events with support staged along the length. It is why long shaft work takes longer to set up here, and why it comes off the machine straight.

> **When this is the wrong machine**
>
> 16" OD ceiling and no meaningful through-bore. Larger diameters go to Big Thicket Express. Short aerospace turned parts are wasted on a 120-inch bed.

Full specifications

Model	Hwacheon Mega 95 ("Rattle & Hum")

Type	Heavy-duty gap-bed CNC lathe — Mega-95 platform
Max turning capacity	16" OD × 120" L (B&R working spec)
Platform max swing over bed	38.5"
Swing through gap	48.4"
Swing over cross slide	26.2"
Distance between centers	10 ft (3,000 mm) B&R config; platform available to 205"
X-axis travel	19.09"
Spindle bore	6"
Spindle speed	40 – 1,000 RPM (heavy-duty range)
Spindle motor	30 HP
Chuck	24" 3-jaw hydraulic
Turret	12-position
Control	Fanuc 21-T (typical)
Machine weight	~22,000 lb
Tolerance capability	±0.0005" routine on critical features; straightness held over full length
Materials	Full oilfield alloy range — Inconel, Duplex, 17-4 PH, 4140/4340, Monel

Values marked "platform" reflect the machine manufacturer's published spec range for this platform family; B&R's working spec is what we quote on this specific unit.

Who this serves in Aerospace

Typical buyer: Aerospace primes, tier-1 and tier-2 suppliers, MRO shops, propulsion component manufacturers, structural assembly integrators, actuator OEMs.

What's hard about aerospace work: Certification-driven documentation, exotic superalloys, tight tolerances on structural and engine components, and finding a job shop that understands aged-condition superalloys rather than learning on the part.

Typical Aerospace parts we produce on Rattle & Hum

- Extended landing gear structural components
- Long engine and gearbox shafts
- Extended aerospace structural rods
- Long-form actuator components
- Aerospace parts requiring 8+ foot turning length

Above list is representative; if your part isn't shown, call and ask — most oilfield / aerospace / defense / industrial turned or milled work fits somewhere in the shop.

Alloys we run for Aerospace

Standard range: Titanium Grade 5 (Ti-6Al-4V) and Grade 23 (Ti-6Al-4V ELI) · Inconel 718 · Inconel 625 · Waspaloy · A286 · 17-4 PH · 15-5 PH · 13-8 Mo PH · Aluminum 7075/6061 · Custom aerospace alloys on request.

Titanium Grade 5 is the workhorse and it's temperature-sensitive: aggressive chip load with flood coolant keeps the heat in the chip, not the workpiece. Light cuts overheat titanium and cook tools; that's why titanium demands a different intuition than steel. Inconel 718 in aerospace-aged condition is harder and less forgiving than the annealed billet stock — most aerospace 718 work assumes aged. 17-4 PH and 15-5 PH are common for structural and actuator components; both are precipitation-hardening stainless with condition-specific machining plans.

> **What "we run these weekly" means:** we stock or reliably source the common alloys in the industry's typical spec range. Sharp tooling, proven feeds and speeds, and process control specific to each alloy family. Your part is not the one we're experimenting on.

How we machine it — our 5-step process

This is the process B&R runs on every job, aerospace or otherwise. The specifics of feeds, speeds, and tooling shift by alloy and machine; the discipline is universal.

1. **Material verification.** Every job starts with material traceability. Heat and lot numbers documented against the mill test report. For customer-supplied material, we verify against the CofC before touching the stock.

2. **First-article layout.** Stock is measured and laid out before any cutting starts. Concentricity, roundness, and squareness of the starting material established; if the stock is out of tolerance we call the customer before wasting time on a bad blank.

3. **Roughing with process control for the alloy.** Feeds, speeds, and coolant strategy set for the specific alloy family — sharp coated carbide (or PCD for high-volume Ti Grade 5), positive rake, high-pressure through-tool coolant on 2507 and Inconel to control cutting temperature. Interrupted cuts get extra attention because they cycle heat into the workpiece.

4. **Finish pass to spec.** Finish pass with a fresh insert, controlled DOC, and measured surface-finish targets. Critical features (bore concentricity, seat grooves, sealing surfaces) are held tighter than most job-shop lathes can offer.

5. **CMM verification + documentation.** Every critical dimension CMM-verified. First-article report generated. Certificate of conformance issued with delivery. If your quality group needs a documentation package beyond that, tell us with the RFQ and we'll be straight about what we can and can't produce in-house.

Tolerance and finish expectations in Aerospace

±0.0005" routine on critical features. Tighter with proper fixture and setup — some aerospace features run ±0.0002" or GD&T-driven cylindricity and position tolerances. CMM verification standard on first articles; process-driven measurement on runs. Surface finish measured, not estimated.

Documentation and quality workflow

First-article layout and CMM verification standard. Material traceability by heat and lot. Certificates of conformance issued with delivery. Proprietary prints handled under NDA — prints stay in-shop and aren't sent to outside vendors. To be clear about what we are not: B&R is not AS9100-registered and not ITAR-registered. If your program requires either, you need a registered shop and we'll say so on the first call rather than after you've placed the order.

Response time and emergency work

AOG (aircraft-on-ground) work prioritized when material is available. Emergency response supported for existing aerospace customers with blanket arrangements or established relationships. Call direct at (936) 291-7827.

How we compare

Compared to a captive aerospace shop: we're faster for one-offs and small runs, more flexible on setup, and we don't require a long approval process. Compared to a lowest-bidder commodity shop: we understand the alloys and we treat first-article inspection as a discipline rather than a formality. Where a registered shop beats us is on programs that require AS9100 flow-down — we're the wrong call for those, and the right call for non-controlled work, prototypes, tooling, and MRO parts where alloy competence and turnaround matter more than a certificate on the wall.

Working with B&R Productions

New customer or existing, the process starts the same way: call (936) 291-7827 or send a print through the quote form. Prints are accepted as PDF, STEP, IGES, DWG, or DXF; if you only have a sample part, we reverse-engineer routinely. NDAs on request, and prints stay in-shop rather than going out to third-party vendors. Note for defense and aerospace buyers: B&R is not ITAR- or AS9100-registered, so export-controlled prints and AS9100 flow-down programs need a registered shop — we'll tell you that on the first call, not after the PO.

Quotes typically come back within 24 hours for standard work, same-day for repeat customers on stocked materials, and near-immediately by phone for emergency and rig-down situations. Lead times are quoted honestly — we don't promise Wednesday and deliver next month.

Frequently asked questions

8 questions covering the Hwacheon Mega 95 platform and Aerospace work. Also indexed as FAQ schema for AI answer engines.

What's the Mega 95's working envelope?

16" OD × 120" L (10 ft) is B&R's working spec. Platform max swing is 38.5" over bed with a 48.4" swing through the gap and 26.2" swing over cross slide.

What kind of parts does the Mega 95 run?

Long shafts, extended mandrels, rig-floor rollers, drive rods, and oil-and-gas structural round parts — anything that doesn't fit on a 4- or 6-foot lathe. The 24" 3-jaw hydraulic chuck accepts big roughing blanks.

What's the spindle bore and speed range?

6" spindle bore, 40–1,000 RPM. 30 HP motor. Control is Fanuc 21-T on our unit; machine weight ~22,000 lb.

Do you serve aerospace primes and tier-1/2 suppliers?

On work that doesn't require a registered QMS, yes — that's the honest boundary. B&R is not AS9100-registered, and where that matters it matters absolutely, so we say it before you spend time on an RFQ: if your program requires AS9100 flow-down, we're the wrong shop. Prototypes, tooling, fixtures, MRO and repair parts, and non-controlled components are where a shop like this is genuinely useful to a prime or a tier-1, and that's the work we'd like to quote.

Can you handle ITAR-controlled aerospace/defense work?

No. ITAR compliance requires registration with the U.S. State Department's DDTC, and B&R does not hold that registration. If your print is ITAR-controlled, you need a registered shop — send it elsewhere and we'd rather tell you now than after you've placed an order. Plenty of aerospace and defense work isn't export-controlled, and that work we quote gladly. Either way, proprietary prints are handled under NDA and stay in-shop.

What tolerances do you hold on aerospace parts?

±0.0005" routine on critical features. First-article layout and CMM verification standard. Surface-finish targets measured to spec.

What aerospace alloys do you run?

Titanium Grade 5 (Ti-6Al-4V), Grade 23 ELI, Inconel 718 and 625, Waspaloy, A286, 17-4 PH, 15-5 PH, aluminum 7075/6061. Custom aerospace alloys on request.

Can you respond to AOG (aircraft-on-ground) work?

Yes, when material is available. Emergency response supported for existing aerospace customers. Newer customers should call to discuss what's realistic for their specific alloy and part.

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Ready to talk about your Aerospace work?

Send us a print or a sample photo and specs. Emergencies: call direct and say "rig-down."

Published by B&R Productions — a precision CNC machining shop in New Waverly, Texas, in business since 1994. Serving oil & gas, aerospace, defense, and industrial customers across Texas and the Gulf Coast.

Written by the B&R Productions team. Machine specs verified against manufacturer data. Alloy and process notes reflect shop practice as of 2026-01-15.